

# VIRTUAL BOOKSTORE DATABASE MANAGEMENT SYSTEM

Normalized 3NF Relational Architecture and Advanced SQL Suite

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Domain: Database Engineering

Stack: MySQL, PostgreSQL, SQLite

## SYSTEM FEATURES

**3NF**

Schema Normalization

**12+**

Relational Tables

**30+**

Complex SQL Queries

**SQL**

Triggers & CTEs

## RELATIONAL SCHEMA & SQL SUITE

Designed a normalized relational database in 3NF supporting bookstore operations including inventory tracking, customer orders, reviews, and transaction auditing.

- Schema DDL: Available in `sql/schema.sql`
- Query Suite: Available in `sql/queries.sql` featuring CTEs, Window Functions, and Joins
- Database Runner: `db_manager.py` for automated schema instantiation